

PAVAN KUMAR ETTA

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PROFESSIONAL SUMMARY

Master's graduate in Computer Information Science (Data Science) from the University of Massachusetts. **3+** years of professional experience as a Software Engineer building scalable backend systems, REST APIs, cloud-native applications, and distributed systems using Python, Go, AWS, Kubernetes, and SQL.

SKILLS

Programming: Python, JavaScript, TypeScript, C/C++, Java, Go

Software Engineering: Data Structures & Algorithms, OOP, Microservices, Distributed Systems

Backend & APIs: Django, Flask, FastAPI, REST APIs, gRPC, GraphQL, Event-Driven Architecture

Frameworks: jQuery, Angular, Node.js, Vue.js, Next.js, Flutter, Ngix, Figma, HTTPS

Databases & Caching: PostgreSQL, MySQL, DynamoDB, MongoDB, Redis, Elasticsearch

Cloud & Infra: AWS (Lambda, EC2, S3, RDS, MSK, EKS), GCP, Azure, Terraform, Serverless

Data Engineering: Apache Spark, PySpark, Airflow, Snowflake, Hadoop, dbt

DevOps & Observability: Docker, Kubernetes, CI/CD, GitHub Actions, GitLab CI

WORK EXPERIENCE

Graduate Research Assistant | University of Massachusetts, Dartmouth | USA

Mar 2025 – Dec 2025

- Led a Next.js-based web application with a Flask and MongoDB as backend, enabling users to upload financial documents and view detailed reports generated through an NLP pipeline.
- Integrated LangChain for AI workflow orchestration and optimized ChatGPT-based text extraction, improving extraction accuracy by **30%**; implemented BERT-based classification with **93%** accuracy.

Software Engineer II | Rhenix Lifesciences LLP | India

Aug 2021 – Aug 2024

- Developed cross-platform CureOrg features enabling patients to connect **5+** wearable platforms and access heart rate, sleep, activity, calories, distance, and elevation data through Fitbit, Garmin, Withings, and Apple Health integrations.
- Automated daily health-data retrieval, reducing collection effort by nearly **90%**, by building Python ingestion jobs and Django commands on **EC2** to manage authentication, refresh tokens, parse JSON, retry failed requests, and store validated records.
- Reduced health-data sync debugging time by around **40%** by scheduling cron-based ingestion jobs and implementing audit logs, reconciliation checks, and AWS CloudWatch alerts to detect failed APIs, delayed executions, and missing records.
- Improved research-data reliability by building a SQL ingestion function to aggregate raw 5-second data readings into 1-minute metrics, removing duplicates by user and timestamp, flag missing data, and upsert validated records into MySQL on AWS EC2.
- Enhanced health-metrics dashboard performance by up to **70%** by optimizing Django REST endpoints responses, MySQL query patterns, indexing, pagination, and Redis caching for frequently accessed user metrics.
- Participated across the Software Development Life Cycle, including requirements analysis, API documentation, system design, implementation, testing, deployment, monitoring, troubleshooting, and production support.

Software Engineer Intern | Cure Science Institute | USA

Jun 2021 – Aug 2021

- Developed a BLE-based asset tracking system using an Android app, BLE beacons, gateways, and REST APIs, enabling real-time location data uploads to AWS S3 with persistent background services for 24/7 tracking and improved battery efficiency.
- Architected admin analytics dashboard using React, Google Maps, AWS API Gateway, DynamoDB & RDS, CloudWatch, and scheduled jobs to monitor device health, visualize coordinates, detect offline gateways, and analyze asset movement patterns.

PROJECTS

Distributed API Gateway & Rate Limiting Service [\[GITHUB\]](#)

(Python, Go, AWS, Redis, Docker, Kubernetes)

- Designed a cloud-native API gateway with authentication, request routing, rate limiting, retries, circuit breakers, and service-level observability, improving backend reliability and protecting microservices from traffic spikes.

Fault-Tolerant Sharded Key-Value Store [\[GITHUB\]](#)

(C++, Raft, SQLite)

- Engineered a fault-tolerant sharded key-value store in C++ using Raft consensus, SQLite, leader election, log replication, and replicated state machines to maintain consistency and support recovery from server failures.

Automated Subjective Answer Evaluation System [\[GITHUB\]](#)

(Flask, BERT, Machine Learning, Docker, AWS EC2/S3)

- Developed an automated answer-evaluation platform using Python, Flask APIs, BERT, Decision Trees, Random Forest, and XGBoost to process more than 10,000 responses and achieve **95%** classification accuracy.

Wandermate [\[GITHUB\]](#)

(React, TypeScript, Node.js, MongoDB, Socket.IO)

- Developed a full-stack travel platform using React, TypeScript, Node.js, and MongoDB, integrating Socket.IO for real-time messaging, PropelAuth for secure authentication, Mapbox for interactive maps, and reusable APIs for third-party integrations.

EDUCATION

University of Massachusetts, Dartmouth | GPA: 4.0

North Dartmouth, USA

M.S. in Computer Information Science, Data Science

Aug 2024 - May 2026

Indian Institute of Information Technology | GPA: 3.2

SriCity, INDIA

B.Tech in Electronics and Communication Engineering

Aug 2017 - Jun 2021